higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

19 April 2023

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1 1.1.1 A body of mass 'm' at a height 'h' above the ground possesses a potential energy 'mgh'.✓✓

1.1.2 Relative humidity is the ratio of the amount of moisture in a gas mixture to the amount of moisture in a saturated mixture at equal volume, temperature and pressure.✓✓

1.1.3 Gain is the output divided by the input.✓✓

1.1.4 Electrodes acquire a certain surface potential due to loss or gain of electrons when an electric current is passed between the electrodes in a solution of electrolysis.✓✓

1.1.5 Stoke's law relates the viscosity of gasses and liquids to the steady-state velocity of a known-weight sphere falling through a fluid of known density at a very low Reynolds number.✓✓

(5 × 2) (10)

1.2 1.2.1 Vena contractor point✓✓
1.2.2 Concentric orifice plate✓✓
1.2.3 Variable area meter/rotameter✓✓
1.2.4 Dew point✓✓
1.2.5 Non-Newtonian fluid✓✓

(5 × 2) (10)

1.3 1.3.1 True
1.3.2 True
1.3.3 True
1.3.4 True
1.3.5 True

(5 × 1) (5)
[25]

TOTAL SECTION A: 25

SECTION B

QUESTION 2: FLOW MEASUREMENTS

2.1 Impact pressure developed:

$H = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$ where $V_2 = 0$ ✓

Therefore: $h = \frac{-V_1^2}{2g}$ ✓

The pressure increases by $\frac{V_1^2}{2g}$.✓

Increase in head $h = \frac{V_1^2}{2g}$ ✓ **OR** $V_1^2 = 2gh$

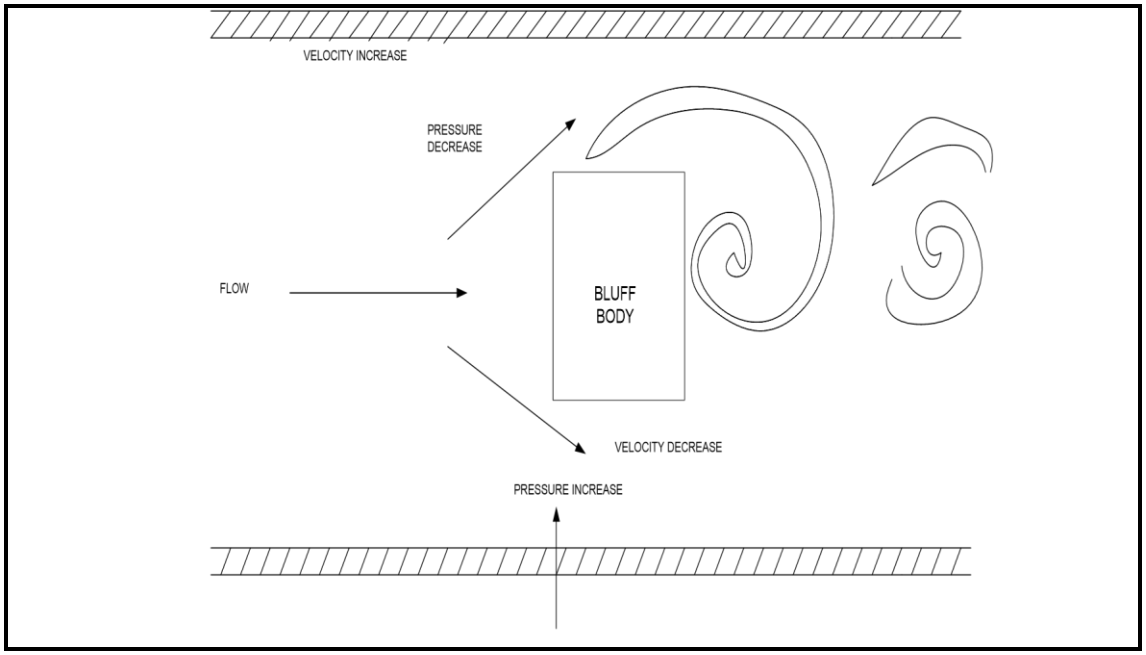
Thus: $V_1 = \sqrt{2gh}$ ✓

Pitot tube coefficient is introduced: $V_1 = c\sqrt{2gh}$ ✓ (6)

2.2 A pitot tube measures point velocity✓ whilst an annubar transverses the flow and measures average velocity.✓ (2)

2.3 Faraday's law states that when a conductor cuts a magnetic field, an emf is induced in the conductor.✓ The magnitude of the emf depends on the strength of the magnetic field, the length of the conductor✓ and the speed with which the conductor cuts through the magnetic field.✓ (3)

2.4



(4)

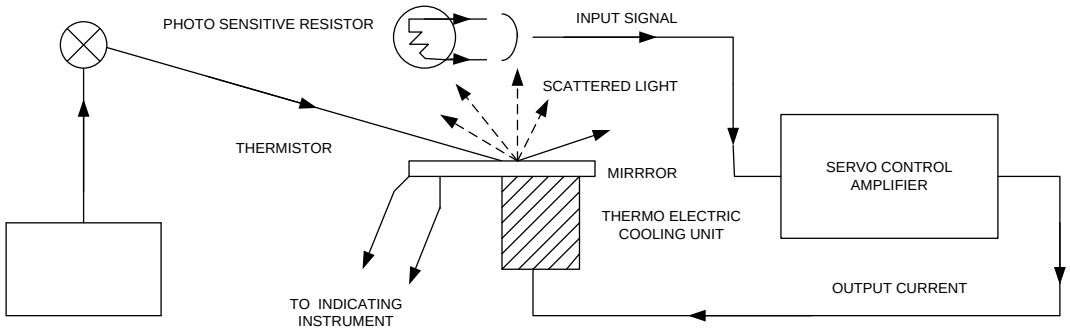
2.5

ADVANTAGES	DISADVANTAGES
• Excellent range ability✓ • No moving parts✓ • With the linear range, density temperature, pressure and viscosity variation have very little effect on it✓ • Very low pressure drop✓ (Any THREE)	• Limited application data✓ • In-line mounting requirement✓ • Limitations imposed on upstream and downstream piping requirement✓ (Any TWO)

(3 + 2) (5)
[20]

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY

3.1



(2 for correct sketch + ½ for each correct label) (7)

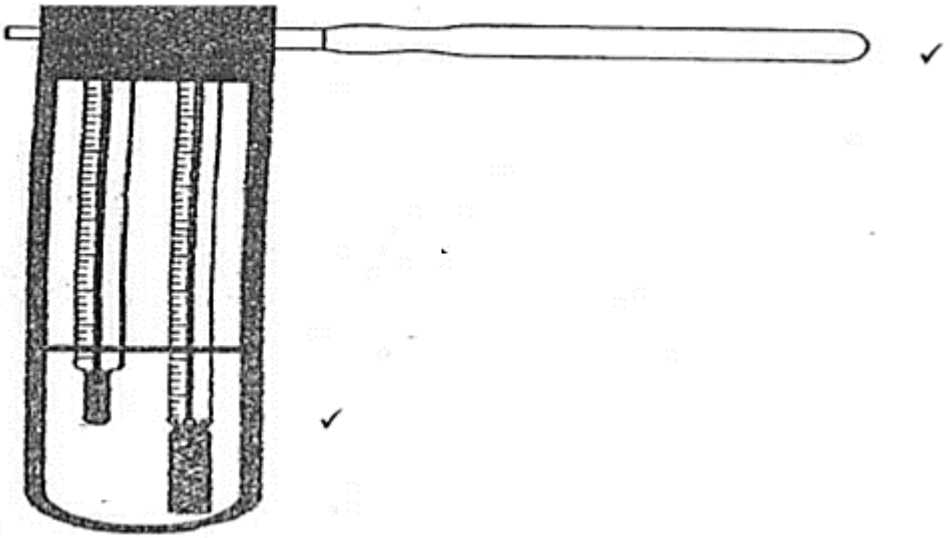
3.2 3.2.1 A: Synchronous motor√
B: Gear train√
C: Dial√
D: Calibrated spring√
E: Bearing and pivot√
F: Spindle coupling nut√
G: Spindle guard√
H: Spindle body√
I: Sample container√
J: Immersion mark√ (10 × ½) (5)

3.2.2 Coaxial-cylinder viscometer✓✓ (2)

3.2.3
- Speed of rotation
- Spindle size and shape
- Temperature of liquid
- Size and shape of sample container
- Depth of liquid from bottom of the container
- Spindle immersion depth
- Elapsed time of the rotation before reading is taken

(Any 5 × 1) (5)

3.3



Two mercury-in-glass thermometers are securely fastened in a frame pivoted on a handle.✓ The wet-bulb assembly with its wick is so constructed that it cannot spill during operation.✓ The frame is rotated by hand.✓ The thermometer readings are noted and the relative humidity must be found from the thermometer tablets.✓ (6)

[25]

QUESTION 4: pH MEASUREMENT

4.1 $pH = -\log[H^+]$

$[H^+] = 10^{-pH}$

$[H^+] = 10^{-4,4}$✓

$= 4 \times 10^{-5}$ mole/dm^{-5}✓

$pOH = 14 - pH$

$= 14 - 4,4$✓

$= 9,6$✓

$[OH] = 10^{-9,6}$✓

$= 2,5 \times 10^{-10}$ mole/dm^3✓ (6)

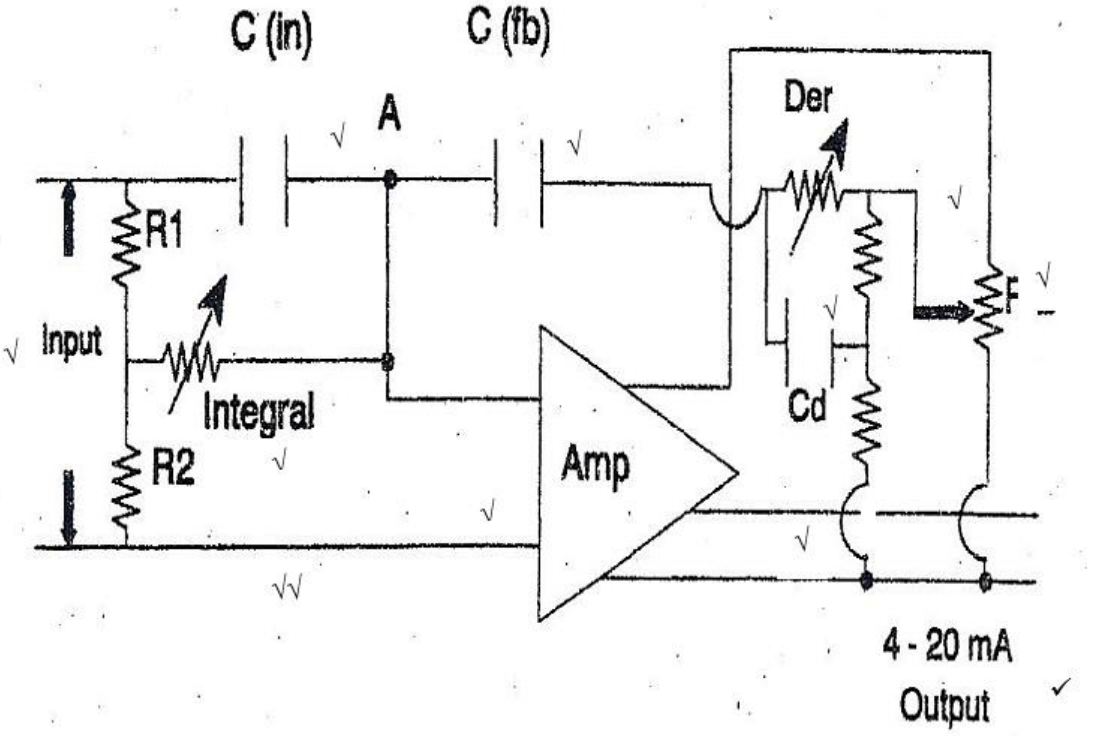
4.2
- Glass electrode
- Hydrogen electrode
- Quinhydrone electrode
- Tungsten electrode
- Platinum electrode
- Iodine electrode (6)

4.3
- Temperature
- Mixture and reaction rates
- Electrode surface condition (3)

[15]

QUESTION 5: AUTOMATIC CONTROL

5.1



(2 for correct sketch + ½ for each labelling) (7)

5.2 Controller sensitivity is the percentage proportional band.✓ The smaller the band, the more sensitive the output.✓ The bigger the band, the slower the output.✓ (3)

5.3 To eliminate the offset caused by proportional control (1)

5.4 Inherent control is a balanced condition in a system without a controller. (1)

5.5 In an open-loop control there is no feedback signal✓ to a controller to reset the process.✓ Any deviation in the process must be corrected by hand.✓ (3)

[15]

TOTALSECTION B: 75
GRAND TOTAL: 100

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